OS Lab 3 — Process Scheduling: Simulator Results and Analysis

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Assignment Summary

We implemented and ran a scheduler simulator for multiple scheduling algorithms and workloads. For each workload (`process1.dat`, `process2.dat`, and `process3.dat`) we tested:

- FIFO (single CPU and 2 CPUs)
- Non-preemptive SJF (single CPU and 2 CPUs)
- Preemptive SJF (single CPU and 2 CPUs)
- Round Robin (single CPU and 2 CPUs) with different quantum values: 2, 4, 6

For each run, the simulator outputs the schedule and the performance metrics: **Simulator Runtime**, **Average Turnaround Time**, and **Maximum Turnaround Time**. We collected those metrics and compared algorithms using both tables and graphs.

Process1.dat — Results

Run	Simulator runtime (ms)	Avg. turnaround	Max. turnaround
FIFO (single CPU)	0.100	696.143	1460
FIFO (2 CPUs)	0.050	533.429	723
NP-SJF (single)	0.094	696.143	1460
NP-SJF (2 CPUs)	0.106	445.714	842
P-SJF (single)	0.232	611.571	1419
P-SJF (2 CPUs)	0.237	354.571	722
RR (single, q=2)	1.083	872.286	1431
RR (2 CPUs, q=2)	1.567	443.000	759
RR (single, q=4)	0.562	878.143	1431
RR (2 CPUs, q=4)	0.781	444.714	760
RR (single, q=6)	0.466	877.143	1429
RR (2 CPUs, q=6)	0.648	445.857	760

Observations:

- With two CPUs, RR performance is much better (average turnaround ≈ 445) regardless of quantum size.
- For single CPU, changing quantum from 2 to 6 did not significantly reduce turnaround — the values stayed high (≈ 872).
- Simulator runtime decreased with larger quantum because the simulator executed fewer scheduling iterations and file writes. Although no real context switch overhead is modeled in the code, this trend qualitatively resembles the theoretical expectation.

Process2.dat — Results

Run	Simulator runtime (ms)	Avg. turnaround	Max. turnaround
FIFO (single CPU)	0.185	165.167	963
FIFO (2 CPUs)	0.048	260.167	482
NP-SJF (single)	0.179	165.167	963
NP-SJF (2 CPUs)	0.223	134.056	581
P-SJF (single)	0.221	161.944	942
P-SJF (2 CPUs)	0.234	92.000	567
RR (single, q=2)	0.804	248.389	936
RR (2 CPUs, q=2)	1.137	122.667	501
RR (single, q=4)	0.412	241.333	933
RR (2 CPUs, q=4)	0.714	119.556	500
RR (single, q=6)	0.288	258.389	935
RR (2 CPUs, q=6)	0.420	129.500	496

Observations:

- The best case was RR with 2 CPUs and q=4, giving 119.56 average turnaround.
- Increasing quantum too much (q=6) slightly worsened turnaround compared to q=4.
- Larger quanta consistently reduced simulator runtime because fewer scheduling iterations were performed. This does not model real CPU context switching costs, but the effect mirrors the theoretical expectation.

Process3.dat — Results

Run	Simulator runtime (ms)	Avg. turnaround	Max. turnaround
FIFO (single CPU)	0.371	1017.500	2199
FIFO (2 CPUs)	0.075	633.000	1061
NP-SJF (single)	0.332	1017.500	2199
NP-SJF (2 CPUs)	0.436	531.167	1166
P-SJF (single)	0.394	931.667	2071
P-SJF (2 CPUs)	0.415	498.667	1050
RR (single, q=2)	1.579	1261.750	2100
RR (2 CPUs, q=2)	2.320	631.167	1054
RR (single, q=4)	0.776	1279.333	2087
RR (2 CPUs, q=4)	1.210	641.000	1050
RR (single, q=6)	0.705	1226.917	2051
RR (2 CPUs, q=6)	0.964	621.667	1041

Observations:

- Preemptive SJF (2 CPUs) was still the best performer overall.
- For Round Robin, two CPUs consistently reduced turnaround to nearly half compared to single CPU.
- For 2 CPUs, increasing quantum from 2 to 6 had little effect on turnaround (values stayed ≈ 630).

Effect of Quantum Size in Round Robin

We now focus only on Round Robin and compare q=2, q=4, and q=6.

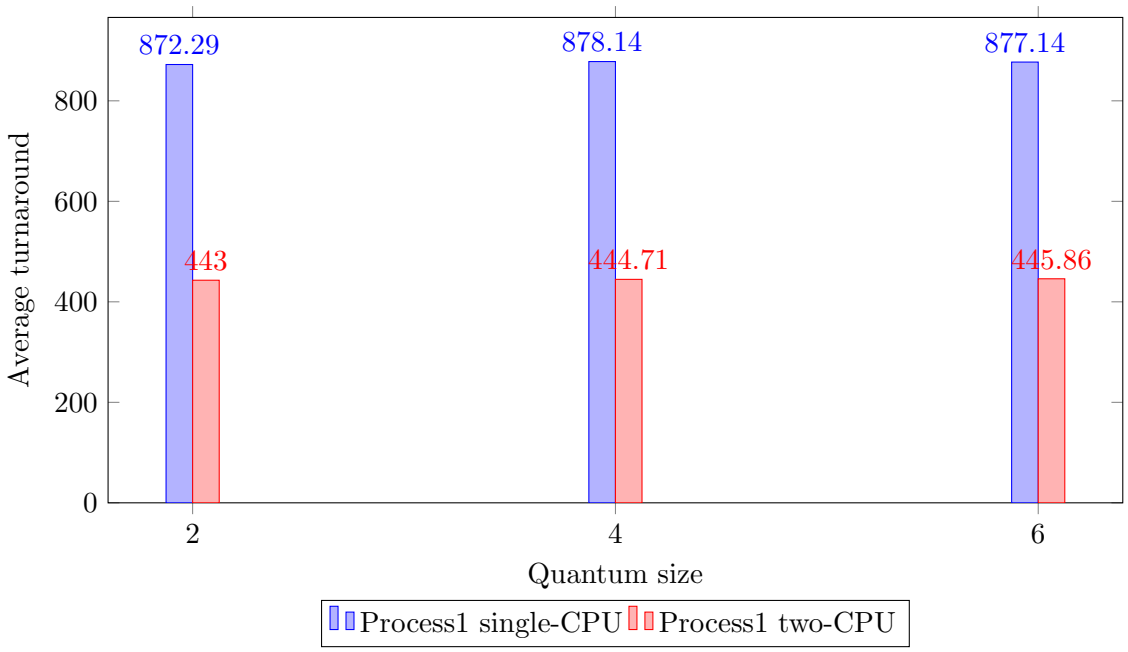


Figure 1: Round Robin avg. turnaround vs quantum (process1).

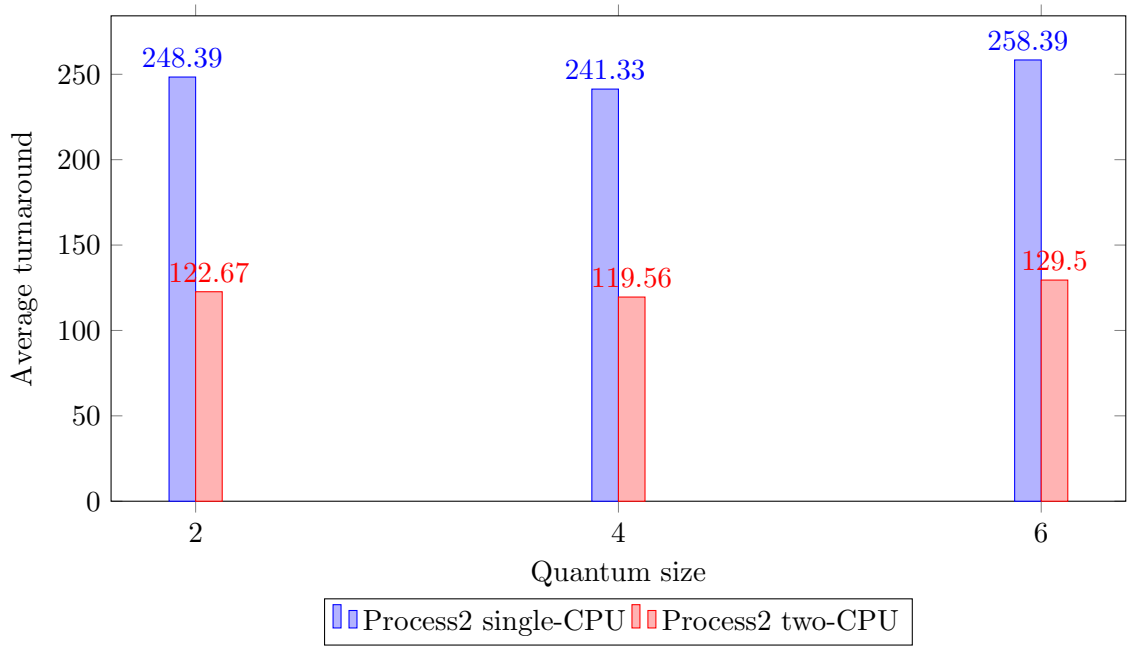


Figure 2: Round Robin avg. turnaround vs quantum (process2).

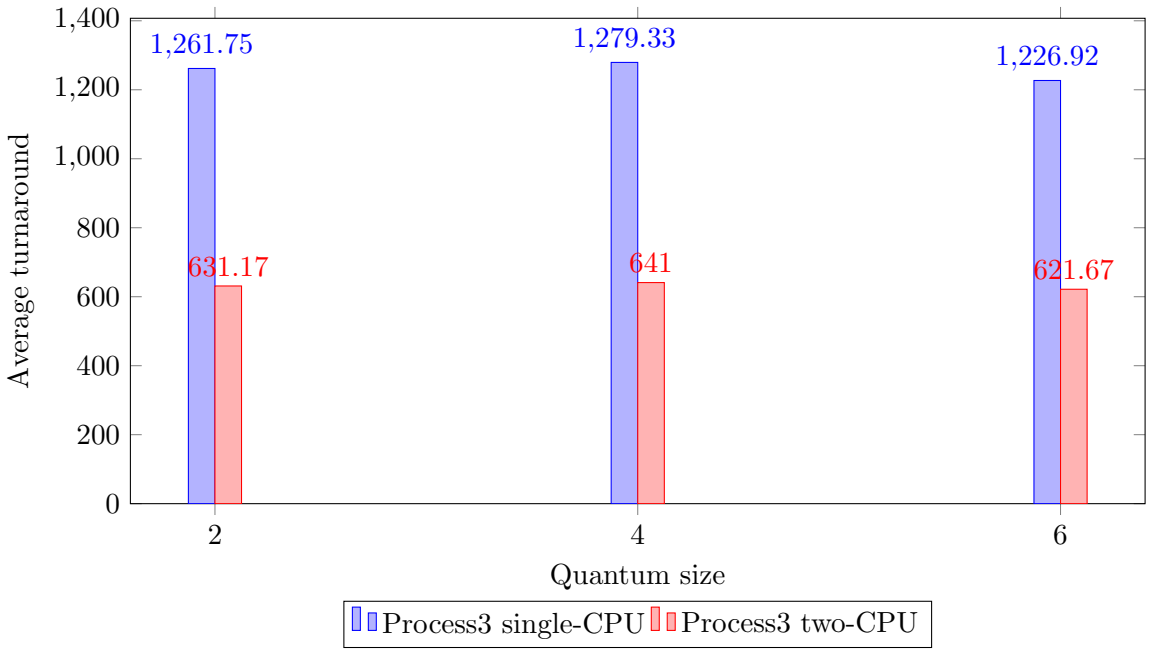


Figure 3: Round Robin avg. turnaround vs quantum (process3).

Simulator Runtime vs Quantum Size (Round Robin)

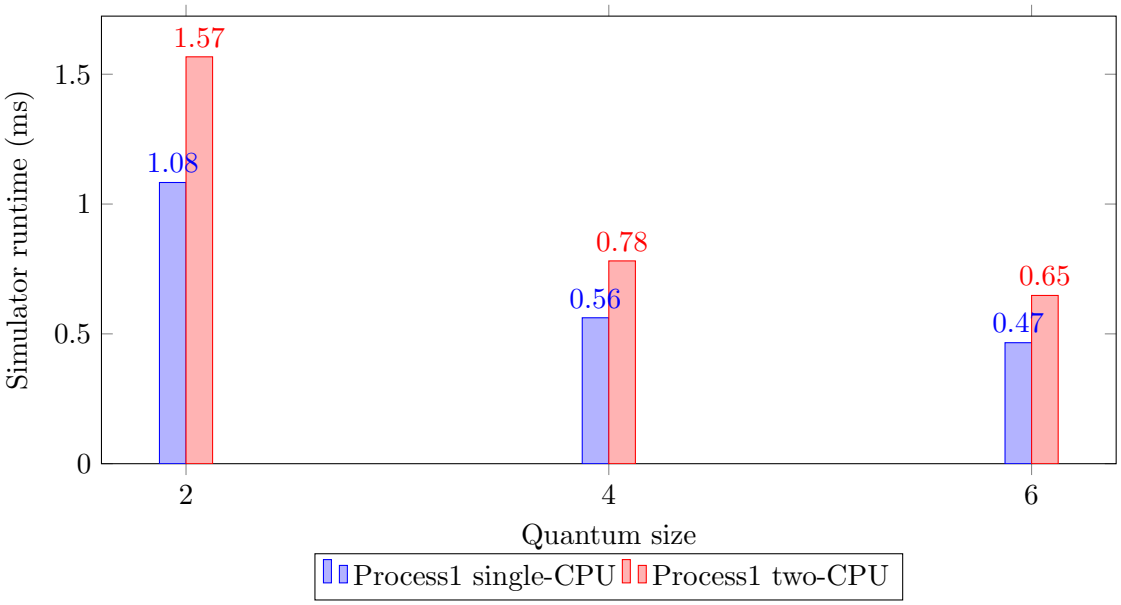


Figure 4: Round Robin simulator runtime vs quantum (process1).

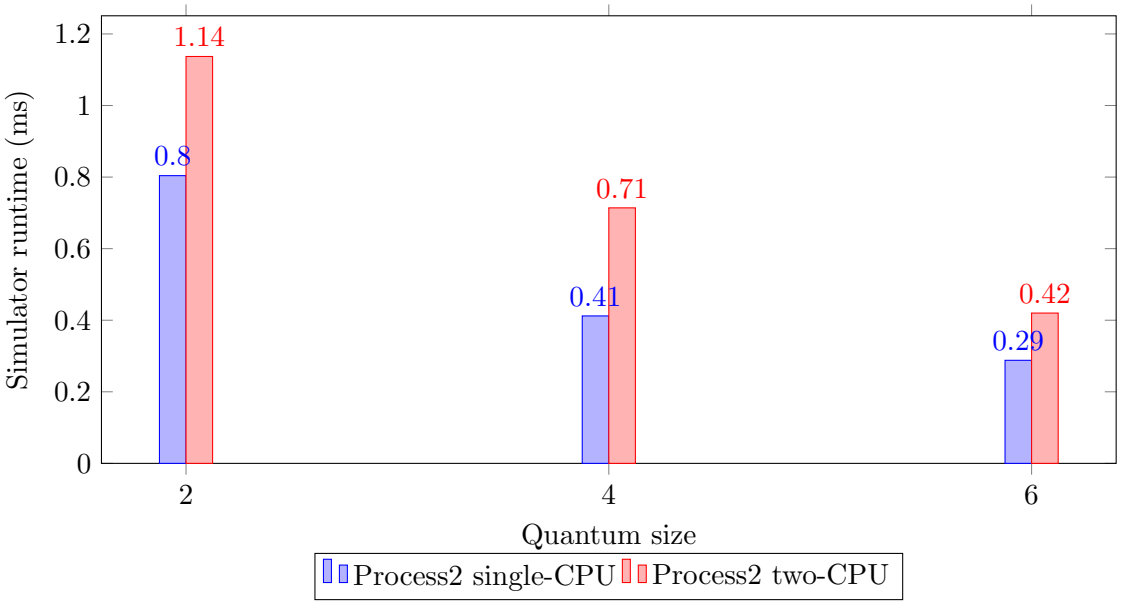


Figure 5: Round Robin simulator runtime vs quantum (process2).

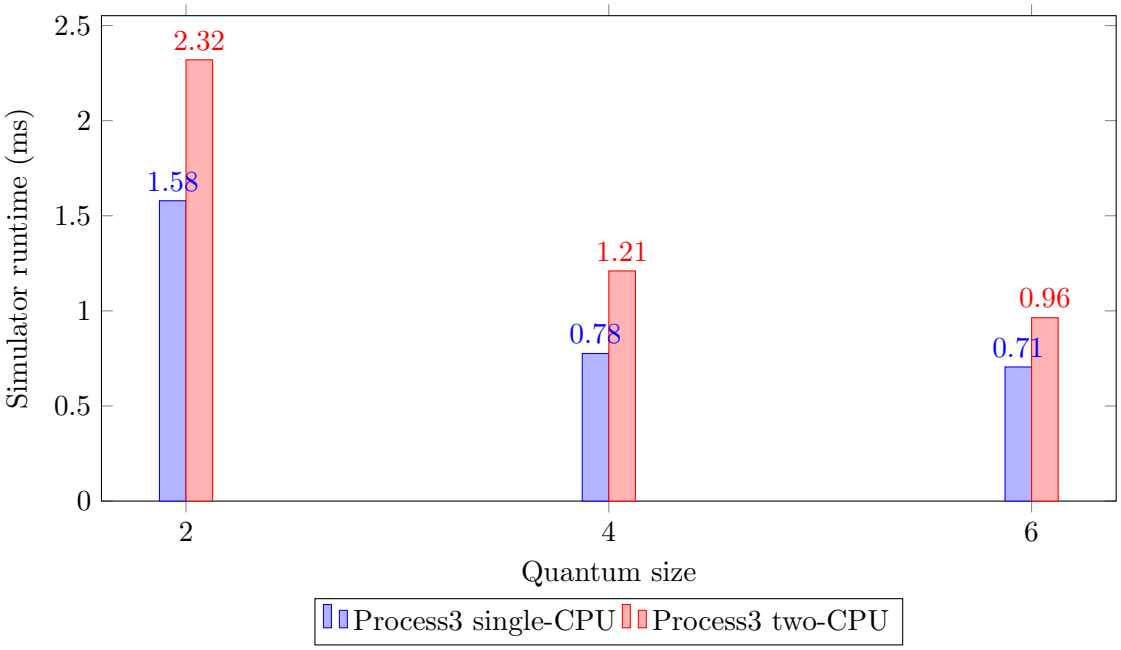


Figure 6: Round Robin simulator runtime vs quantum (process3).

Observation: Simulator runtime dropped consistently as quantum increased, because the program executed fewer scheduling iterations and file writes. This is not actual context switch overhead, but the trend is qualitatively similar to real systems.

Conclusion

The experiments broadly confirmed theoretical expectations, with measured data supporting each point:

- **Shortest Job First (SJF)** minimized turnaround time. For example, preemptive SJF with two CPUs achieved an average turnaround of **498.667** (process3), compared to FIFO's **1017.500**.
- **FIFO scheduling** was simple but inefficient, giving the worst maximum turnaround (**2199** on process3).
- **Round Robin (RR)** balanced fairness but showed limited sensitivity to larger quantum. For process3, averages were **1261.750** ($q = 2$), **1279.333** ($q = 4$), and **1226.917** ($q = 6$). Runtime decreased with larger quanta due to fewer simulator iterations, not actual context switch overhead.
- **Multiple CPUs** consistently improved results. With RR on process3 ($q = 2$), average turnaround dropped from **1261.750** (single CPU) to **631.167** (two CPUs).

In summary: SJF (especially preemptive) gave the lowest turnaround times, FIFO was consistently inefficient, RR balanced fairness but quantum changes had little effect beyond simulator runtime, and multi-CPU scheduling gave clear benefits across algorithms.